DS5020 Mini Project

Individual Reflection

Aditya Vishwanath

What I did

I completed this project on my own, holding both the Data Engineer and the ML Engineer roles. I set up the GitHub repository, the access token and a Google Colab workflow linked to Google Drive, then ran every step myself, checked each output, and fixed the errors along the way: a parquet type error, a dropped Drive connection and a repository rename. On the data side, I cleaned 1.07 million invoice lines down to 776,577, built a time-based train/test split, and packaged the pipeline into a single script that rebuilds the matrix identically. On the ML side, I compared PCA with Truncated SVD and evaluated recommendations against a best-seller baseline. I also built a six-segment customer map and wrote the memo and slides for the stakeholder.

What I learned

The most useful lesson was that the first result can be wrong for a reason you can explain. My first model lost to the best sellers (23.6% vs 26.9%), because cosine similarity throws away how popular a product is. Switching to dot-product scoring raised the hit rate to 40%. I also learned that small data issues matter: a trailing space in one stock code, ‘47503J ‘, splits a single product into two. On the practical side, I got a good recap of how Git, tokens and reproducible pipelines fit together, and why keeping customer IDs out of shared outputs needs active checking rather than assumptions.

What I would change

I would choose k and the scoring method on a separate validation period instead of the test months, which makes my result slightly optimistic. That would make me more independent and better prepared to explain my choices.